

Thermal arithmetic orbit weights and the shifted-zeta equation

Bost–Connes coefficients and the causal realization problem

Drafted for Edward Baker

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Abstract

The arithmetic orbit coefficients in the shifted completed-zeta quotient admit an exact thermal description. In the Bost–Connes system, a fixed primitive projection P_n has equilibrium expectation $\varphi_\beta(P_n) = \prod_{\ell|n} (1 - \ell^{-\beta})$. With inverse temperature $\beta = 2\omega$, the required coefficient is $c_n(\omega) = n^{(\beta-1)/2} \varphi_\beta(P_n)$. This law reproduces primes, repetitions, composites and the prime part of the shift generator, but does not by itself construct a physical response. We derive local positive models and complete two coupling tests. A primitive-controlled delay network has an ordinary energy balance and an identity limit, but fails the arithmetic repetition law. Separately, the unchanged archimedean completion times any finite Euler product has, for $0 < \beta < 1$, an uncanceled pole incompatible with passivity. Native energy-changing probes give a positive Kubo admittance coupled to the full modular half-shift channel. Any such parallel load with finite total response spectral mass preserves the original front exponent and cannot generate the required shift or logarithmic tangent. These scoped exclusions leave the coefficient identity intact and motivate direct cumulative-storage and canonical-system investigations. A causal completed physical realization remains open.

Status. Exact coefficient matching; causal completed physical realization open; regular finite-place parallel-load route parked. The 87-, 144- and 203-case diagnostic records, their limitations and LLM assistance are documented in [Appendix A](#).

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1 Thermal orbit weights: the current result

The current investigation seeks an independently defined physical system whose response supplies the arithmetic side of the shifted-zeta differential equation. Its most useful new result concerns the changing orbit weights. The Bost–Connes system has native projections that detect primitiveness at specified primes. Their positive thermal expectations, with one common inverse temperature, give precisely the product entering every required arithmetic coefficient. No separate choice is made for each prime or each repeated orbit.

More precisely, let ℓ always denote a prime, let φ_β be a Kubo–Martin–Schwinger (KMS) equilibrium state at inverse temperature β , and let P_n select states primitive at every prime dividing n . The dictionary is

$$\boxed{\beta = 2\omega, \quad c_n(\omega) = n^{(\beta-1)/2} \varphi_\beta(P_n), \quad \varphi_\beta(P_n) = \prod_{\ell|n} (1 - \ell^{-\beta}).} \quad (1.1)$$

The primitive projector is independent of temperature. The displayed centering factor is additional and explicit; its physical origin is not claimed to follow merely from the existence of a thermal state.

The established Bost–Connes theory supplies positive KMS states in the range $0 < \beta \leq 1$, although the usual global Gibbs trace does not converge there [1, 2]. This is the range needed to move from the modular half shift $\omega = 1/2$ toward zero. Thus the candidate is more informative than a formal identification of a zeta partition function. A finite primitive observable has a meaningful positive expectation at the relevant temperatures.

The contribution of this investigation is the exact comparison with the desired orbit coefficients and the analysis of specific readout and norm requirements. The underlying equilibrium system, KMS existence theorem and primitive-count identities are established mathematics. There is no claim of literature novelty for them.

Three levels must remain separate. First, the primitive probabilities in (1.1) are native thermal observables. Second, their centered values reproduce the Euler coefficients, including the entire prime part of the shift generator. Third, a causal physical response with these amplitudes, the archimedean completion and the ordinary radiation norm has not been constructed. Positivity at the first level does not prove the third.

The completed interface tests sharpen this distinction. A controlled unitary delay model realizes actual logarithmic flight times, an ordinary energy balance and an identity limit, but has the wrong arithmetic amplitudes. Moreover, the unchanged archimedean factor times a finite Euler product cannot be globally passive. A subsequent energy-exchanging thermal load retains the full modular core and is passive as a linear boundary system, yet fails a structural front-exponent test. These results justify parking this regular boundary-load branch, not rejecting the equilibrium coefficient identity or every possible global coupling.

Sections 2–3 establish the dictionary, and Section 4 gives the local norms and assembly limitations. Section 5 presents the completed finite-place test and completion obstruction; Section 6 gives the energy-exchanging response and finite-spectral-mass theorem. Section 7 summarizes earlier tests, and Section 8 states the remaining priorities. The principal research sources are the orbit-weight note [9] and the two subsequent test notes [6, 7]. Their numerical records and same-assistant audits are distinguished from the analytical arguments in Appendix A.

2 The arithmetic response and its orbit coefficients

2.1 The shift equation and the physical requirement

Define the completed function and its centered logarithmic derivative by

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad H(p) = \xi(\tfrac{1}{2} + p), \quad m(p) = \frac{H'(p)}{H(p)}. \quad (2.1)$$

The target transfer satisfies

$$K_\omega(p) = \frac{H(p-\omega)}{H(p+\omega)}, \quad \partial_\omega K_\omega = -a_\omega K_\omega, \quad a_\omega(p) = m(p-\omega) + m(p+\omega), \quad K_0 = 1. \quad (2.2)$$

These identities are initially used on a sufficiently far right, zero-free half-plane, and then meromorphically. Here p is the Laplace variable, ω the arithmetic shift, and β the thermal parameter in (1.1); physical time is a separate variable.

A physical realization must specify its states, preparation, propagation and readout, and derive K_ω from them. In particular, on an ordinary radiation space $L^2(0, L)$ its causal compression $V_{\omega, L}$ must have the appropriate identity limit and accumulated norm balance. The desired contraction is

$$\|V_{\omega, L}f\|_2^2 \leq \|f\|_2^2, \quad V_{0, L} = I. \quad (2.3)$$

The strong identity limit is the relevant convention; an operator-norm derivative at zero is not presumed. Cumulative contraction does not require every shift increment to be contractive, or a_ω itself to be a holomorphic positive-real function. Those distinctions and the unconditional weighted bounds are developed in [12].

Writing down the quotient in (2.2) is not a proof of the full unweighted realization. Its unit boundary modulus, where defined, follows from the functional equation and conjugation; causal contractivity for every positive shift is a further issue, related to the RH-equivalent innerness criterion [3]. No such conclusion is inferred here from the thermal coefficient identity.

2.2 Euler coefficients at unchanged logarithmic labels

For $\text{Re } p > 1/2 + \omega$, factor the target as

$$K_\omega(p) = A_\omega(p)Z_\omega(p), \quad Z_\omega(p) = \frac{\zeta(p + 1/2 - \omega)}{\zeta(p + 1/2 + \omega)}, \quad (2.4)$$

$$A_\omega(p) = \pi^\omega \frac{\Gamma(s_-/2)}{\Gamma(s_+/2)} \frac{s_-(s_- - 1)}{s_+(s_+ - 1)}, \quad s_\pm = p + \tfrac{1}{2} \pm \omega. \quad (2.5)$$

For one prime, set $a = \ell^{\omega-1/2}$, $b = \ell^{-\omega-1/2}$ and $z = \ell^{-p}$. Then

$$\frac{1 - bz}{1 - az} = 1 + \sum_{r \geq 1} (1 - \ell^{-2\omega}) \ell^{r(\omega-1/2)} z^r. \quad (2.6)$$

Multiplying the absolutely convergent local series gives

$$Z_\omega(p) = \sum_{n \geq 1} c_n(\omega) e^{-p \log n}, \quad c_n(\omega) = n^{\omega-1/2} \prod_{\ell|n} (1 - \ell^{-2\omega}), \quad c_1 = 1. \quad (2.7)$$

At $\omega = 1/2$, this is $c_n = \varphi_E(n)/n$, with φ_E Euler's totient. The orbit terminology refers to the primitive residues of the modular constant term, not to the closed-geodesic coefficients of a Selberg

trace formula. The labels $\log n$ stay fixed as the arithmetic weights vary. Whether they are physical delays in a new model must be derived separately.

The first-prime coefficient and its tangent are

$$c_2(\omega) = \frac{2^\omega - 2^{-\omega}}{\sqrt{2}}, \quad c_2(1/2) = \frac{1}{2}, \quad c'_2(1/2) = \frac{3}{2} \log 2. \quad (2.8)$$

The full logarithmic derivative also follows in the same convergence region:

$$\partial_\omega \log Z_\omega(p) = 2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \cosh(\omega \log n) e^{-p \log n} = -a_{\omega, \text{prime}}(p). \quad (2.9)$$

The transfer involves all positive integers through multiplicativity; its logarithmic source involves prime powers through the von Mangoldt function Λ . The remaining terms of $-a_\omega$ are $\partial_\omega \log A_\omega$. A successful physical deformation must produce those terms with the same parameter.

3 Primitive projections in the Bost–Connes system

3.1 The equilibrium identity

The Bost–Connes arithmetic system was introduced in [1]. We use its multiplicative isometries v_n and the range projections $e_n = v_n v_n^*$, with

$$v_n^* v_n = I, \quad v_m v_n = v_{mn}, \quad e_m e_n = e_{\text{lcm}(m,n)}, \quad \sigma_t(v_n) = n^{it} v_n. \quad (3.1)$$

In the standard integer representation $v_n |k\rangle = |nk\rangle$, the projection e_n detects divisibility by n , and the Hamiltonian is $\mathcal{H}|k\rangle = (\log k)|k\rangle$. This representation has a normalized Gibbs trace only for $\beta > 1$. The full system nevertheless has positive KMS states for $0 < \beta \leq 1$; their existence and local measure structure are established inputs, for which we use [2, Introduction and Section 2].

For analytic elements the KMS condition reads $\varphi_\beta(ab) = \varphi_\beta(b\sigma_{i\beta}(a))$. Apply it to $a = v_n$ and $b = v_n^*$:

$$\varphi_\beta(e_n) = \varphi_\beta(v_n^* \sigma_{i\beta}(v_n)) = n^{-\beta}. \quad (3.2)$$

No analytic continuation of a divergent trace is used in this identity.

Proposition 3.1 (Primitive thermal coefficient identity). *For $n \geq 1$ define the temperature-independent projection*

$$P_n = \prod_{\ell|n} (I - e_\ell), \quad P_1 = I. \quad (3.3)$$

For a KMS state as above,

$$\varphi_\beta(P_n) = \prod_{\ell|n} (1 - \ell^{-\beta}). \quad (3.4)$$

Consequently the centered expectations at $\beta = 2\omega$ are exactly the coefficients in (2.7).

Proof. The divisibility projections commute. Expanding the finite product in (3.3) and using the lcm relation gives

$$P_n = \sum_{d|\text{rad}(n)} \mu(d) e_d, \quad (3.5)$$

where $\text{rad}(n)$ is the product of the distinct prime divisors, μ is the Möbius function and $e_1 = I$. Taking expectations and applying (3.2) gives the product in (3.4). Multiplication by $n^{(\beta-1)/2}$ gives (2.7). Thus the result does not require a separate probabilistic independence assumption. $\square$

At a repeated prime, $P_{\ell^r} = P_\ell$: primitiveness is the same event. Its contribution is

$$c_{\ell^r}(\omega) = (1 - \ell^{-2\omega})\ell^{r(\omega-1/2)}. \quad (3.6)$$

The repetition dependence comes from centering; one has not introduced a new primitive event for each r . For coprime integers the projections combine by their distinct prime sets, recovering the required multiplicativity. Hence the first-prime tangent (2.8) and the full prime source (2.9) follow from the same coefficient law.

There is also a direct counting interpretation at integer exponent. Inclusion-exclusion shows that

$$J_d(n) = n^d \prod_{\ell|n} (1 - \ell^{-d}) \quad (3.7)$$

counts tuples $(x_1, \dots, x_d)$ modulo n with $\gcd(n, x_1, \dots, x_d) = 1$. For real $\beta > 0$ the analogous product defines $J_\beta(n)$, and $c_n(\omega) = J_{2\omega}(n)/n^{\omega+1/2}$. The thermal construction supplies a real-parameter probability law; it does not posit a noninteger number of discrete coordinates.

3.2 An explicit positive model at finitely many places

Let S be a finite set of primes. On $\bigotimes_{\ell \in S} \ell^2(\mathbb{N}_0)$, with occupation operators N_ℓ , take the self-adjoint Hamiltonian

$$\mathcal{H}_S = \sum_{\ell \in S} (\log \ell) N_\ell. \quad (3.8)$$

For every $\beta > 0$ its Gibbs state is trace class, with

$$\mathcal{Z}_S(\beta) = \prod_{\ell \in S} (1 - \ell^{-\beta})^{-1}, \quad \Pr_\beta(N_\ell = m) = (1 - \ell^{-\beta})\ell^{-m\beta}. \quad (3.9)$$

The primitive projection selects occupation zero at each prime dividing n , provided those primes belong to S . It gives (3.4) directly. This finite-place realization of the relevant diagonal observables is explicit and positive throughout the temperature range of interest. It is not asserted to encode every observable of the full arithmetic algebra.

As a small illustration, the centered coefficients are

n	$\omega = 1/4$	$\omega = 0.4$	$\omega = 1/2$
2	0.246292858	0.397146260	1/2
3	0.321144348	0.523917402	2/3
4	0.207106781	0.370550563	1/2
6	0.079095559	0.208071837	1/3

The entries are samples of the exact identity, not a fit defining the state. In the infinite system for $\beta \leq 1$, the product partition function diverges. The positive KMS state must then be used in its established algebraic sense, not replaced by $e^{-\beta\mathcal{H}}/\zeta(\beta)$.

3.3 What the centering and the time variable still require

For $0 < \beta \leq 1$, the centered observable

$$B_{n,\beta} = n^{(\beta-1)/2} P_n \quad (3.10)$$

is a positive contraction and $\varphi_\beta(B_{n,\beta}) = c_n(\beta/2)$. However, this observable depends on temperature. The equality alone does not derive all amplitudes from a temperature change with a fixed physical detector.

The dynamics gives a structured analytic-time expression:

$$c_n(\beta/2) = n^{-1/2} \varphi_\beta \left(P_n v_n^* \sigma_{-i\beta/2}(v_n) \right), \quad \sigma_{-i\beta/2}(v_n) = n^{\beta/2} v_n. \quad (3.11)$$

It identifies a thermal half-step and the fixed factor $n^{-1/2}$. Imaginary-time continuation is not unitary real-time propagation. The physical input/output normalization that would implement (3.11) is still to be derived.

There is a second, independent distinction. Since $\sigma_t(P_n) = P_n$, a native real-time correlation of two such projections is constant:

$$\varphi_\beta(P_m \sigma_t(P_n)) = \varphi_\beta(P_m P_n). \quad (3.12)$$

The number $\log n$ in $\sigma_t(v_n) = n^{it} v_n$ is an energy difference or dilation label. It has not thereby become the causal delay $\log n$ in (2.7). That conversion is a requirement on a coupling to propagating signals, not a consequence of equilibrium positivity.

4 Ordinary local norms and two assembly limitations

4.1 A radial realization of the diagonal state

For one prime ℓ , normalize additive Haar measure dx on $\mathbb{Q}_\ell$ by $\text{vol}(\mathbb{Z}_\ell) = 1$ and define

$$d\mu_{\beta,\ell}(x) = C_{\beta,\ell} |x|_\ell^{\beta-1} dx, \quad C_{\beta,\ell} = \frac{1 - \ell^{-\beta}}{1 - \ell^{-1}}. \quad (4.1)$$

This is the local scaling law of the equilibrium diagonal [2, Section 2]. The shell $v_\ell(x) = m$ has Haar mass $(1 - \ell^{-1})\ell^{-m}$. Consequently its mass under (4.1), for $m \geq 0$, is $(1 - \ell^{-\beta})\ell^{-m\beta}$, exactly (3.9). Thus

$$\mu_{\beta,\ell}(\mathbb{Z}_\ell) = 1, \quad \mu_{\beta,\ell}(aE) = |a|_\ell^\beta \mu_{\beta,\ell}(E), \quad \mu_{\beta,\ell}(\ell^r \mathbb{Z}_\ell) = \ell^{-r\beta} \quad (r \geq 0). \quad (4.2)$$

The unit shell supplies the primitive probability $1 - \ell^{-\beta}$.

The normalized half-density in the fixed Haar Hilbert space is

$$u_{\beta,\ell}(x) = \sqrt{C_{\beta,\ell}} |x|_\ell^{(\beta-1)/2} 1_{\mathbb{Z}_\ell}(x), \quad \|u_{\beta,\ell}\|_2 = 1. \quad (4.3)$$

This realizes the diagonal state in an ordinary positive norm. It is not claimed to intertwine the full Bost–Connes representations at different temperatures.

Weighted dilations on $L^2(\mu_{\beta,\ell})$ and the canonical half-density transport are

$$D_a^{(\beta)} f(x) = |a|_\ell^{\beta/2} f(ax), \quad T_\beta f(x) = \sqrt{C_{\beta,\ell}} |x|_\ell^{(\beta-1)/2} f(x). \quad (4.4)$$

The scaling law makes $D_a^{(\beta)}$ unitary, and cancellation of the weights gives

$$T_\beta D_a^{(\beta)} T_\beta^{-1} = D_a^{(1)}. \quad (4.5)$$

Therefore a pure measure change does not alter the unitary dilation action after this canonical transport. The parameter enters the state and its observables.

Direct shell integration gives an actual native correlation:

$$\langle u_{\beta,\ell}, D_{\ell^r}^{(1)} u_{\beta,\ell} \rangle = \ell^{-\beta|r|/2}, \quad r \in \mathbb{Z}. \quad (4.6)$$

Indeed the common support has valuation $m \geq \max(0, -r)$ and the integrand summed over shells is $(1 - \ell^{-\beta})\ell^{-\beta m - r\beta/2}$. Its geometric sum proves (4.6). This is different from c_{ℓ^r} : primitive projection and centering remain additional readout choices, even though the local state norm is known.

4.2 Failure of the naive common product vacuum

At the reference temperature $\beta = 1$, $u_{1,\ell} = 1_{\mathbb{Z}_\ell}$. The affinity of the two normalized local vectors is

$$h_\ell(\beta) = \langle u_{1,\ell}, u_{\beta,\ell} \rangle = \frac{\sqrt{(1 - \ell^{-1})(1 - \ell^{-\beta})}}{1 - \ell^{-(\beta+1)/2}}. \quad (4.7)$$

In the incomplete tensor product based on the reference vectors, embed a finite set of changed local states as

$$\Omega_{\beta,S} = \bigotimes_{\ell \in S} u_{\beta,\ell} \otimes \bigotimes_{\ell \notin S} u_{1,\ell}. \quad (4.8)$$

Proposition 4.1 (A scoped product-space obstruction). *For each fixed $0 < \beta < 1$, the finite-place vectors (4.8) are not Cauchy as S exhausts the primes. Their shift tangents at $\beta = 2\omega = 1$ also have unbounded norm.*

Proof. For fixed $\beta < 1$, expansion of (4.7) at large ℓ gives

$$1 - h_\ell(\beta) \sim \frac{1}{2} \ell^{-\beta}. \quad (4.9)$$

The prime sum diverges, since $\ell^{-\beta} \geq \ell^{-1}$ and $\sum \ell^{-1}$ diverges. Hence $\prod_\ell h_\ell(\beta) = 0$. For $S \subset T$ the normalized vectors satisfy

$$\|\Omega_{\beta,T} - \Omega_{\beta,S}\|^2 = 2 \left(1 - \prod_{\ell \in T \setminus S} h_\ell(\beta) \right). \quad (4.10)$$

For every fixed finite S , the right side approaches 2 as T grows, which excludes the Cauchy property.

For the tangent, express the normalized local vector in its shell basis as $a_m(\beta) = \sqrt{(1-q)q^m}$, where $q = \ell^{-\beta}$. Differentiation gives

$$\partial_\beta a_m = \frac{\log \ell}{2} \left(\frac{q}{1-q} - m \right) a_m. \quad (4.11)$$

The geometric variance is $q/(1-q)^2$. Taking account of $\partial_\omega = 2\partial_\beta$ at $\beta = 1$ yields

$$\left\| \partial_\omega \Omega_{2\omega,S} \Big|_{\omega=1/2} \right\|^2 = \sum_{\ell \in S} (\log \ell)^2 \frac{\ell}{(\ell-1)^2}. \quad (4.12)$$

Each local derivative is orthogonal to its normalized real vector, so the product derivative has no cross terms. The last prime sum diverges. $\square$

Some finite-place values illustrate the calculation:

Primes	Largest	Overlap, $\beta = .5$	Overlap, $\beta = .8$	Tangent norm ²
4	7	0.805625	0.978404	3.41186
25	97	0.398762	0.924811	10.9944
168	997	0.0444597	0.819804	23.9650
1,229	9,973	$8.01090 \cdot 10^{-5}$	0.658092	42.3681
9,592	99,991	$1.73891 \cdot 10^{-12}$	0.454913	66.1381

The proof is the divergent-series argument, not the finite table.

This obstruction concerns the stated Haar-based embedding. It does not invalidate the established positive KMS states, whose representations are supplied by their own construction. Nor does it exclude a different common representation, a correlated archimedean coupling, or a different radiation map. Finite-observable thermal expectations can be differentiable even when this vector family is not. No claim about the canonical Sonin embeddings studied elsewhere in the project follows from this calculation.

4.3 The bare Euler delay train is not contractive

A second tempting assembly would turn the numbers $\varphi_\beta(B_{n,\beta})$ directly into delayed scalar amplitudes:

$$R_\omega = I + \sum_{n \geq 2} c_n(\omega) S_{\log n}, \quad (4.13)$$

where $S_v f(u) = f(u - v)$ with zero extension. Each finite observation window sees only finitely many shifts, so the following exclusion requires no definition of the infinite sum as a global bounded operator.

Proposition 4.2 (Bare Euler readout exclusion). *For $\omega > 0$, the compression of (4.13) to any window $\log 2 < L < \log 3$ is not a contraction in ordinary $L^2(0, L)$.*

Proof. Only $I + c_2 S_{\log 2}$ appears on this window. Choose a unit pulse supported on $(0, \varepsilon)$, where $0 < \varepsilon < \min(\log 2, L - \log 2)$. The direct and delayed copies are disjoint and contained in the window. Thus

$$\left\| 1_{(0,L)} R_\omega f \right\|_2^2 = 1 + c_2(\omega)^2 > 1. \quad (4.14)$$

□

The squared norm is approximately 1.06066017177982 at $\omega = 1/4$, and 1.25 at $\omega = 1/2$. Positive contractive observables at individual labels do not control the energy of their coherent sum.

Equation (4.14) does not exclude the completed response $A_\omega Z_\omega$. The same bare Euler failure occurs at the known modular half shift, where the archimedean factor is essential [15]. The next physical test must therefore include that sector and its flux, rather than infer passivity from the signs or individual sizes of the arithmetic weights.

5 The finite-place interface: completed tests

The proposed two-prime interface has now been tested in an explicit unitary delay model. Its ordinary energy balance and identity limit hold, but its arithmetic amplitudes fail. A separate argument excludes a literal finite Euler product with the unchanged archimedean completion. This section summarizes the derivations in [6]; that note also contains the detailed path calculations and numerical controls.

5.1 Primitive-controlled loops and measured amplitudes

Use the finite Gibbs sector (3.8) with $S = \{2, 3\}$ and $q_\ell = \ell^{-\beta}$. At each prime attach a unit-speed transport loop with coordinate $x = \log y$, $1 \leq y \leq \ell$. Its length and flight time are $\tau_\ell = \log \ell$. This geometry is an explicit modeling assumption; a thermal energy label alone does not derive a flight time. In the primitive sector $P_\ell = 1$, impose

$$\begin{pmatrix} y_\ell \\ u_\ell \end{pmatrix} = \begin{pmatrix} -r_\ell & t_\ell \\ t_\ell & r_\ell \end{pmatrix} \begin{pmatrix} f_\ell \\ v_\ell \end{pmatrix}, \quad t_\ell = \sqrt{1 - r_\ell^2}, \quad v_\ell(t) = u_\ell(t - \tau_\ell), \quad r_\ell = \ell^{-1}. \quad (5.1)$$

The complementary sector bypasses an empty, disconnected loop. The full controlled vertex is unitary and preserves the thermal occupations. Prepare the loops empty, cascade the stages, and measure the mean outgoing amplitude. The controller is not reset on a repeated traversal.

Writing $z_\ell = e^{-p\tau_\ell}$, elimination of the internal fields gives

$$F_\ell = \frac{z_\ell - r_\ell}{1 - r_\ell z_\ell}, \quad R_{\ell,\beta} = q_\ell + (1 - q_\ell)F_\ell, \quad R_\beta = R_{2,\beta}R_{3,\beta}. \quad (5.2)$$

Each F_ℓ is a causal contraction, and the product follows from the independent, unchanged controllers. Set $d_\ell = q_\ell - r_\ell(1 - q_\ell)$ and $b_\ell = (1 - q_\ell)(1 - r_\ell^2)$. The coefficients of the paths labeled 1, 2, 3, 4, 6 are respectively

$$g_1 = d_2 d_3, \quad g_2 = b_2 d_3, \quad g_3 = d_2 b_3, \quad g_4 = r_2 b_2 d_3, \quad g_6 = b_2 b_3. \quad (5.3)$$

The primitive probability occurs once per distinct prime, but

$$\frac{g_4}{g_2} = \frac{1}{2}, \quad \frac{c_4(\beta/2)}{c_2(\beta/2)} = 2^{(\beta-1)/2}. \quad (5.4)$$

These ratios differ throughout $0 < \beta \leq 1$. More generally, a fixed loop reflectivity cannot match the varying ratio on an open temperature interval. Formally normalizing by $g_1 > 0$ gives, at $\beta = 1$,

Label n	g_n/g_1	Required $c_n(1/2)$
2	3/2	1/2
3	16/3	2/3
4	3/4	1/2
6	8	1/3

This normalization compares path weights; it does not define another contractive readout. Its first-prime shift tangent is $12 \log 2$, whereas (2.8) requires $(3/2) \log 2$. Moreover, $R_\beta(p) \rightarrow d_2 d_3 > 0$ as real $p \rightarrow +\infty$, while the completed target tends to zero like $(2\pi/p)^{\beta/2}$.

5.2 Energy accounting and a genuine identity limit

For each controller sector a , unitarity of (5.1) gives input energy equal to output energy plus loop storage. Let y_a be the sector output, $\bar{y} = \mathbb{E}_\beta y_a = R_\beta f$, and $E_a(L)$ the total energy remaining in the two loops. Averaging and separating the output variance yields

$$\|f\|_{(0,L)}^2 = \|\bar{y}\|_{(0,L)}^2 + \mathbb{E}_\beta \|y_a - \bar{y}\|_{(0,L)}^2 + \mathbb{E}_\beta E_a(L). \quad (5.5)$$

All terms use ordinary radiation energy. The variance is outgoing signal fluctuation correlated with the controller, not unexplained absorption; the controller occupations do not change. This identity establishes passivity of the specified model, not arithmetic matching.

The model also has the positive endpoint property

$$\|R_\beta - I\| \leq 2[(1 - 2^{-\beta}) + (1 - 3^{-\beta})] \longrightarrow 0 \quad (\beta \downarrow 0). \quad (5.6)$$

Thus a scattering identity limit can exist even when the Gibbs states have no trace-class limit. This does not repair the amplitude failure.

5.3 A fixed bounded thermal readout cannot supply centering

Proposition 5.1 (Finite-place readout limitation). *Fix a finite prime set S , its native Gibbs state ρ_β , and a bounded temperature-independent operator B . For any $n > 1$ supported on S , $\text{Tr}(\rho_\beta B)$ cannot equal $c_n(\beta/2)$ on a nonempty open interval of positive temperatures.*

Proof. In the occupation basis the expectation is

$$f_B(\beta) = \prod_{\ell \in S} (1 - \ell^{-\beta}) \sum_{\mathbf{m} \geq 0} \langle \mathbf{m} | B | \mathbf{m} \rangle e^{-\beta \sum_{\ell} m_{\ell} \log \ell}. \quad (5.7)$$

Bounded diagonal entries give locally uniform convergence on $\operatorname{Re} \beta > 0$. Both f_B and $c_n(\beta/2)$ are holomorphic there, so equality on the interval forces equality throughout this half-plane. For real $\beta > 0$, however, $|f_B(\beta)| \leq \|B\|$, whereas $n^{(\beta-1)/2} \prod_{\ell|n} (1 - \ell^{-\beta}) \rightarrow \infty$ as $\beta \rightarrow \infty$. $\square$

The proposition requires extraction by a fixed bounded observable in the native Gibbs state. It does not apply automatically to coefficients obtained by temperature-dependent deconvolution of a completed response, to a different interacting equilibrium, or to the full $K_{\beta/2}$.

5.4 Why a finite Euler completion is not passive

For a finite prime set S , put

$$Z_{S,\beta/2}(p) = \prod_{\ell \in S} \frac{1 - \ell^{-s_+}}{1 - \ell^{-s_-}}, \quad d = \frac{1 - \beta}{2}, \quad s_- = p + d, \quad s_+ = p + d + \beta. \quad (5.8)$$

At $p = d$, one has $s_+ = 1$ and $s_- = 1 - \beta$. For $0 < \beta < 1$, the rational completion in (2.5) has a simple pole, while every finite Euler factor and both gamma factors are finite and nonzero. Direct evaluation gives

$$\operatorname{Res}_{p=d} [A_{\beta/2} Z_{S,\beta/2}] = -\beta(1 - \beta) \pi^{(\beta-1)/2} \Gamma\left(\frac{1 - \beta}{2}\right) \prod_{\ell \in S} \frac{1 - \ell^{-1}}{1 - \ell^{-(1-\beta)}} \neq 0. \quad (5.9)$$

A causal time-invariant contraction on ordinary $L^2(0, \infty)$ has a holomorphic contractive multiplier in $\operatorname{Re} p > 0$. Hence this finite completion cannot be such a passive response. Additional zero-input passive ports do not help: their measured scalar block is still contractive. The pole allows a causal unstable response, so the exclusion concerns causality together with the required energy bound.

The full quotient cancels the pole through $1/\zeta(s_+)$, using the simple residue-one pole of zeta at one [5]. In fact

$$K_{\beta/2}(d) = 2\xi(1 - \beta). \quad (5.10)$$

This local cancellation is unconditional; it does not establish full-family passivity. At $\beta = 1$, the two-prime completion instead behaves as $-2/(3 \log 2 \log 3 p^3)$ near $p = 0$, whereas $K_{1/2}(0) = 1$. Thus the existing global modular benchmark is unaffected.

Finite *time* windows remain valid comparisons. The far-right Euler series and the causal archimedean kernel imply equality with the full formal causal kernel before the first omitted prime delay [6, Section 7]. For $S = \{2, 3\}$ this means $L < \log 5$. A scalar comparison through $\log 6$ must also include prime 5; a path-resolved two-loop experiment can isolate label 6. Finite-window agreement alone asserts neither passivity nor a physical realization.

6 Energy-exchanging boundary loads and front rigidity

A second test allows the thermal sector to exchange energy and connects its response to the *full* modular channel. It yields a positive linear boundary model but a structural arithmetic exclusion. Complete derivations, additional endpoint checks and the diagnostic record are in [7].

6.1 Native Kubo response and its physical scope

Write $H_S = \sum_{\ell=2,3} \epsilon_\ell N_\ell$, with $\epsilon_\ell = \log \ell$, and retain the native Gibbs state. The isometry $v_\ell|m\rangle = |m+1\rangle$ obeys $v_\ell^* v_\ell = I$ and $v_\ell v_\ell^* = I - P_\ell$. Unlike the primitive gate, the bounded self-adjoint probe $B_\ell = v_\ell + v_\ell^*$ changes the occupation energy. A real boundary effort drives the interaction

$$H_{\text{probe}}(t) = H_S - U(t) \sum_{\ell} g_\ell B_\ell. \quad (6.1)$$

For bounded real U , this is a bounded self-adjoint perturbation on the original domain. All dynamical times use the common radiation time conjugate to p ; inverse temperature is a separate parameter.

The retarded response around the uncoupled Gibbs state is obtained from the Kubo commutator [4]. Since $[B_\ell(t), B_\ell(0)] = -2i \sin(\epsilon_\ell t) P_\ell$,

$$\chi_{\ell,\beta}(t) = 2(1 - q_\ell) \sin(\epsilon_\ell t) 1_{t>0}, \quad \chi_{\ell,\beta}(p) = \frac{2\epsilon_\ell(1 - q_\ell)}{p^2 + \epsilon_\ell^2}. \quad (6.2)$$

The conjugate flow $I = \sum_{\ell} g_\ell d\langle B_\ell \rangle / dt$ has admittance

$$Y_\beta(p) = \sum_{\ell=2,3} w_{\ell,\beta} \frac{p}{p^2 + \epsilon_\ell^2}, \quad w_{\ell,\beta} = 2g_\ell^2 \epsilon_\ell (1 - q_\ell) > 0. \quad (6.3)$$

It is positive real in $\mathbb{H}_R$, because $p/(p^2 + \epsilon^2) = \frac{1}{2}[(p - i\epsilon)^{-1} + (p + i\epsilon)^{-1}]$. Here $\log \ell$ is a transition frequency, not a derived flight delay.

For a common strength $\lambda \geq 0$, this susceptibility has an exact conservative linear realization:

$$\ddot{x}_\ell + \epsilon_\ell^2 x_\ell = \sqrt{\lambda w_{\ell,\beta}} U, \quad I = \sum_{\ell} \sqrt{\lambda w_{\ell,\beta}} \dot{x}_\ell, \quad (6.4)$$

$$E_{\text{load}} = \frac{1}{2} \sum_{\ell} (|\dot{x}_\ell|^2 + \epsilon_\ell^2 |x_\ell|^2), \quad \dot{E}_{\text{load}} = \text{Re}(\bar{U} I). \quad (6.5)$$

The initial coherent perturbation is zero. Equation (6.2) is the exact susceptibility of the uncoupled probe; (6.5) is an exact *linear realization* of it. Neither statement identifies the closed-loop model with exact finite-coupling microscopic quantum scattering. That would require an interacting equilibrium, radiative dressing and fluctuation channels. Indeed

$$\langle B_\ell(t) B_\ell(0) \rangle_\beta = e^{-i\epsilon_\ell t} + q_\ell e^{i\epsilon_\ell t}. \quad (6.6)$$

The load energy balance concerns coherent perturbations, not the full thermal output intensity of a noiseless quantum device.

6.2 Connection to the completed modular response

Use the established half-shift reference $K(p) = \xi(p)/\xi(p+1)$ and $r_b(p) = e^{-bp} K(p)$ at an exterior port. Its fixed-shift causality is an input from the known innerness result and modular construction [3, 15]. The favorable control $b = 0$ removes the extra lead algebraically; it is not asserted to be the selected regular geometric cross-section.

In the fixed measured-output orientation, let f, g be external amplitudes and $a, r_b a$ the core amplitudes. Impose common effort and addition of flows:

$$U = f - g = a - r_b a, \quad f + g = a + r_b a + I, \quad I = \lambda Y_\beta U. \quad (6.7)$$

Then $f = a + I/2$, $g = r_b a + I/2$, and elimination yields

$$S_{\beta,\lambda,b} = \frac{r_b + J_\beta(1 - r_b)}{1 + J_\beta(1 - r_b)}, \quad J_\beta = \lambda Y_\beta/2. \quad (6.8)$$

The entire completed modular transfer is retained. The core admittance $(1 + r_b)/(1 - r_b)$ is positive real, so adding λY_β and taking its Cayley transform proves that (6.8) is analytic and contractive in $\mathbb{H}_R$. For its numerator N and denominator D , the flux identity is

$$|D|^2 - |N|^2 = 1 - |r_b|^2 + \lambda \operatorname{Re} Y_\beta |1 - r_b|^2 \geq 0. \quad (6.9)$$

The ordinary finite-window coherent balance is

$$\|f\|_{(0,L)}^2 - \|g\|_{(0,L)}^2 = E_{\text{load}}(L) + \|a\|_{(0,L)}^2 - \|r_b a\|_{(0,L)}^2 \geq 0. \quad (6.10)$$

Here $r_b a$ denotes causal convolution. The core deficit is the later output energy for its input stopped at L . This uses the established half-shift isometry, not positivity of the unknown variable-shift target.

6.3 A structural exclusion for finite spectral mass

Proposition 6.1 (Fixed-core parallel-load front rigidity). *Suppose $K(p) \sim c p^{-1/2}$, $c > 0$, on the positive real axis, and a load has the representation*

$$Y(p) = \int_{(0,\infty)} \frac{p}{p^2 + \Omega^2} d\nu(\Omega), \quad \nu \geq 0, \quad C := \nu((0,\infty)) < \infty. \quad (6.11)$$

For each fixed finite $\lambda \geq 0$, the connection (6.8) at $b = 0$ satisfies

$$S(p) = K(p) + \frac{\lambda C}{2p} + o(p^{-1}), \quad S(p) \sim c p^{-1/2}. \quad (6.12)$$

For the modular core, it cannot equal $K_{\beta/2}$ at any $0 < \beta < 1$.

Proof. Dominated convergence gives $pY(p) = \int p^2/(p^2 + \Omega^2) d\nu(\Omega) \rightarrow C$. Subtracting the core from (6.8) gives

$$S - K = \frac{(\lambda Y/2)(1 - K)^2}{1 + (\lambda Y/2)(1 - K)}. \quad (6.13)$$

This proves (6.12). The required leading behavior is $K_{\beta/2}(p) \sim (2\pi/p)^{\beta/2}$, which has a different power for $0 < \beta < 1$. $\square$

The hypothesis includes countably many modes when their total response spectral mass is finite. For a self-adjoint thermal probe B , the transition measure has positive weights $2(E_n - E_m)(\rho_m - \rho_n)|B_{mn}|^2$ at $E_n - E_m > 0$. When the sum converges its total mass is

$$C_\beta = \operatorname{Tr} \rho_\beta[B, [H_S, B]]. \quad (6.14)$$

Finite sums of products of the native isometries and their adjoints have bounded energy jumps and the needed bounded commutators. In particular, (6.3) has $C_\beta = \sum_\ell w_{\ell,\beta} < \infty$, without truncating the occupation space. Boundedness of an arbitrary B alone does not imply this energy regularity for an unbounded Hamiltonian.

At a regular exterior port $b > 0$, a nonzero load instead gives $S_{\beta,\lambda,b}(p) \sim \lambda C_\beta/(2p)$: it introduces a prompt response before the reference delay of $e^{-bp}K_{\beta/2}$. Thus the regular port does not repair the failure already visible in the favorable $b = 0$ comparison.

6.4 The calibrated tangent and endpoints also fail

A fixed active added load changes the half-shift reference itself. Grant the favorable additional switch $\lambda(\beta) = 1 - \beta$ so that $S_{1,0,0} = K$. This switch is not supplied by the KMS law. Its microscopic amplitude is proportional to $\sqrt{1 - \beta}$; a differentiable amplitude vanishing at one would give zero first response tangent. The one-sided tangent under the granted switch is

$$T_{\text{native}}(p) = \partial_{\beta} \log S_{\beta,1-\beta,0}(p)|_1 = -\frac{Y_1(p)(1 - K(p))^2}{2K(p)} \sim -\frac{C_1}{2\sqrt{2\pi}}p^{-1/2}, \quad (6.15)$$

$$T_{\text{arith}}(p) = \partial_{\beta} \log K_{\beta/2}(p)|_1 = -\frac{1}{2}a_{1/2}(p) = -\frac{1}{2} \log \frac{p}{2\pi} + O(p^{-1}). \quad (6.16)$$

For $g_2 = g_3 = 1$, $C_1 = \log 2 + (4/3) \log 3$. No constant calibration can turn the decaying native tangent into the required logarithm. In time, the native transfer tangent has a finite step at the front, whereas the arithmetic transfer tangent has a $t^{-1/2} \log t$ singularity [7, Section 7].

For the explicit finite-transition load, $Y_{\beta}(p) = O(p)$ near zero, so $S'_{\beta,\lambda,0}(0) = K'(0)$, while the target slope varies with β . For bounded $\lambda(\beta)$, the native weights vanish as $\beta \downarrow 0$, giving $S \rightarrow K$ pointwise rather than the identity. These supplementary failures are not needed for the front-rigidity proof.

The resulting decision is to park the regular finite-place passive parallel-load route. The theorem concerns the unchanged core and (6.11); it excludes neither distributed interactions that change the core nor every passive boundary admittance. A singular continuum proposal would need its own well-defined domain and positive energy. The coefficient identity in (1.1) remains intact.

7 Earlier tests: a brief account of the exclusions

The earlier investigations narrowed the required physical readout. Their exclusions concern the specified constructions, rather than entire field theories or all possible couplings. The detailed calculations, sources and diagnostics remain in the cited notes.

WZW evolution and the reflected collar. The deterministic SU(2) level-2 pilot gave an explicit selected block and a positive integrated tensor-norm balance for a constant Loewner driver. General moving drivers did not preserve contraction, and the raw smeared kernel did not approach the required identity operator [10]. A subsequent full-module collar had a positive norm balance and a strong identity limit, but its fixed reflected readout was self-adjoint. A bounded self-adjoint causal scalar operator on an interval is a multiplication operator, which excludes the nontrivial arithmetic Volterra response. The selected primary also had the wrong gamma-tower weights and no first-prime delay [11].

Direct arithmetic Loewner sources. The completed xi logarithmic derivative generated genuine normalized Loewner evolutions after a safe spectral translation, with an exact formula recovering the arithmetic source. The arithmetic information was supplied through xi itself. Removing the safe translation did not follow from the geometric construction or its weighted bound, so this did not furnish an independently motivated physical readout [12].

Brownian bridge and first-passage readouts. Bridge-range moments supplied an independent scalar xi identity. The specified positive first-passage readout was causal and had an ordinary norm balance, but its Laplace response was not the required quotient: its kernel had the wrong front

behavior and no arithmetic singularity at $\log 2$. A logarithmic readout recovered the boundary phase without establishing its full-family causality. The distinction was between the native moment identity and the measured response [13, 14].

Modular scattering and a cusp load. Passing to exact one-forms supplied the rational completion of modular scattering and realized $K_{1/2}(p) = \xi(p)/\xi(p+1)$ with the stated ordinary radiation norm and fixed-shift causality. This remains a positive benchmark. The tested real, frequency-independent cusp delta load was lossless and causal but had the wrong arithmetic shift tangent, so it did not extend that benchmark to the required family [15].

Fractional clocks and scalar holonomy. A positive weighted diffusion field varied the leading front exponent, but its scalar clock composition with the modular response failed the subleading term, boundary modulus, fixed prime delays and identity endpoint. An analytical exclusion covered scalar complete-Bernstein clocks, not general coherent couplings [16]. A flat scalar unitary twist on the original modular surface also supplies no continuous control: $S^2 = R^3 = 1$ allows only six characters, and trivial cusp holonomy leaves only the trivial character. Nontrivial cusp holonomy removes the open constant mode [9].

These earlier tests motivated changing primitive orbit weights and the archimedean response together. Sections 5 and 6 carry that proposal through two explicit thermal-interface tests. Their failures concern the stated readouts and fixed-core boundary architecture, while the half-shift benchmark and thermal coefficient identity remain available.

8 Remaining avenues

The completed tests narrow the thermal realization program while leaving the main arithmetic positivity question open. The following priorities update the earlier research assessment [8] in light of [6, 7].

1. **Cumulative storage for the complete transfer.** The next concrete calculation is an ordinary L^2 energy balance on $\log 2 < L < \log 3$. In this window the formal kernel contains the full archimedean term and its first translated copy, so the first arithmetic interference terms can be treated exactly. The goal is a positive storage formula from independently specified states and dynamics. Defining storage to be the desired norm deficit would not prove its positivity. No such formula is established here.
2. **Explicit canonical systems from arithmetic data.** A complementary route seeks a positive Hamiltonian directly from the prime and gamma data without assuming the target's contractivity. Suzuki's explicit unconditional construction has the restriction $\omega > 1$ [3]; extension into the required small-shift range is an unresolved step, not an available consequence of that result.
3. **Global deformations of modular scattering.** The exact-one-form half-shift channel remains a useful fixed reference. An independently motivated deformation of its asymptotic propagation or bulk structure could lie outside Proposition 6.1. Its first tests should be the varying front exponent and logarithmic tangent, before arithmetic echoes are fitted. This is exploratory; no qualifying deformation has been constructed.
4. **Causality and support tests.** Contour, residue and support analyses can isolate what prevents an unweighted causal realization and keep finite-window calculations faithful to the

completed target. Such diagnostics support the storage and canonical-system work; they do not by themselves establish the required positive norm.

5. **Independent review of the retained results.** The thermal coefficient identity, finite-place completion obstruction and finite-spectral-mass theorem merit specialist review with their hypotheses kept explicit. Further loop fitting or adding regular boundary modes within the excluded class is not a priority.

A Numerical evidence, provenance and acknowledgements

A.1 The orbit-weight diagnostic

The orbit-weight investigation has an existing record of 87 passing controls: 17 exact integer cases and 70 floating diagnostics. The manuscript integrates those results; it does not claim a new experiment or numerical proof. Its program and record are linked below, relative to the manuscript directory:

- [Orbit-weight diagnostic program](#).
- [87-case numerical record](#).
- [Same-assistant mathematical and numerical audit](#).

The exact controls enumerate primitive residue tuples in dimensions 1, 2 and 3 for moduli 2, 3, 4 and 6, check divisibility-projection products, and enumerate scalar character phases. These finite checks do not replace the general algebraic arguments in the text.

The floating controls compare the Euler products with a Möbius convolution, a full zeta quotient with its truncated Dirichlet series, and a differentiated finite Euler product with its prime-power source. They also check finite-place state masses, local half-density norms, affinities, native dilation correlations, shift tangents, and the piecewise pulse-energy calculation. The affinity table in Section 4 retains only compact samples, not a large generated dataset.

The recorded environment is Python 3.10.0 with mpmath 1.3.0 at 50 decimal digits. Shell sums run through occupation 360, the Dirichlet comparison through $n = 1500$, and the prime table through 100,000. For the comparison at the positive real Laplace parameter $p = 8$, the elementary bound $0 < c_n(\omega) \leq n^{\omega-1/2}$ gives, with $\alpha = p + 1/2 - \omega > 1$,

$$0 \leq Z_\omega(p) - \sum_{n \leq N} c_n(\omega) n^{-p} \leq \frac{N^{1-\alpha}}{\alpha - 1}. \quad (\text{A.1})$$

Evaluating this proven bound in floating arithmetic does not make the comparison an interval certificate. No xi-zero table or external dataset is used. The record stores a SHA-256 program hash, which establishes file identity rather than correctness.

A.2 The two completed interface diagnostics

The finite-place interface record contains 144 passing floating controls, and the thermal boundary record contains 203. They are separate from the 87-case coefficient suite:

- [Finite-place interface program](#), [144-case record](#), and [research audit](#).
- [Thermal boundary program](#), [203-case record](#), and [research audit](#).

Both use Python 3.10.0 and mpmath 1.3.0 at 60 decimal digits. The first derives echo paths from the boundary recursion, solves the vertex equations, checks coherent/fluctuating/stored energy, and samples the completion residues and their cancellation in the full quotient. Pulse intersection lengths are exact formulas evaluated in floating arithmetic. The second evaluates native correlations and Kubo response, checks the spectral sum rule and integrated oscillator work, independently solves the port equations, and checks passivity, asymptotics, tangents and endpoints. Occupation sums extend through 360 and 400 respectively, with omitted-mass bounds; the second uses infinite-ladder matrix elements, not an artificially truncated commutator.

Passing controls reproduce both the valid energy balances and the failures of arithmetic matching. None is an interval certificate, a proof of a general asymptotic theorem, or a simulation of exact finite-coupling microscopic quantum scattering. The recorded program hashes and case outcomes were checked for this manuscript integration; no numerical program or record was changed and no suite was rerun merely for the editorial revision.

A.3 Historical diagnostics

Earlier suites are separate: the WZW pilot has 72 controls, the direct arithmetic source 60, the collar 60, Brownian readout 51, modular scattering 65, and the fractional cusp test 80. Their notes document their exact scopes. The two earlier general-shift quadrature implementations were corrected and fully replayed during the modular test [15, Section 8]; the analytical exclusions were unchanged. Those historical suites have not been rerun merely for this editorial revision.

A.4 Research and document provenance

The principal sources are [9, 6, 7]. The earlier notes are cited in Section 7 so that the retrospective need not repeat their derivations. The local README and build record identify the active TeX inputs, delivered PDF, source notes, numerical records and review. The manuscript compiles from local TeX inputs; it does not import the parent Wilson–Loewner manuscript. The milestone index records this live revision without creating a dated manuscript snapshot.

The coefficient proofs and scoped exclusions are analytical arguments. The numerical controls do not prove the existence of a causal physical interface, the full-family unweighted contraction, or RH. The completed interface tests narrow the candidate class; the next proposed arithmetic calculation is cumulative storage on the first prime window in Section 8.

A.5 Acknowledgements and review status

This manuscript was drafted for Edward Baker with substantial LLM assistance from GPT-6 (Codex, according to the developer-provided model identity), including mathematical synthesis, algebraic calculations, diagnostic programming, source checking and document preparation. The reasoning-effort setting was not exposed and is not inferred. The research and manuscript audits were performed by the same assistant; no independent specialist review is claimed. Established results are attributed to their sources, and the unresolved centering, causality, completion and norm requirements are stated explicitly.

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